

James Lewis (S2 - 29)

Assignment 2

Module 2: Advanced Python

Course Outcome 2 - Use features of files, directories and regular expression in python for file manipulation.

```
In [31]: import re
```

Expression to match the required pattern

```
In [32]: def extract_critical_errors(log_data: str) -> list[tuple]:
# Regular expression to match the required pattern
log_pattern = re.compile(
    r'\[(?P<timestamp>\d{4}-\d{2}-\d{2} \d{2}:\d{2}:\d{2})\] '
    r'\[ERROR\] '
    r'\[(?P<module>[\w_]+\)] '
    r'(?P<message>.*(?:?:25[0-5]|2[0-4]\d|1\d{2}|[1-9]?d)(?:\.(?:25[0-5]|2[0-4]\d|1\d{2}|[1-9]?d))?(?:25[0-5]|2[0-4]\d|1\d{2}|[1-9]?d)(?:\.(?:25[0-5]|2[0-4]\d|1\d{2}|[1-9]?d))?(?:25[0-5]|2[0-4]\d|1\d{2}|[1-9]?d)(?:\.(?:25[0-5]|2[0-4]\d|1\d{2}|[1-9]?d))?'
)

# Extract relevant log entries
results = []
for match in log_pattern.finditer(log_data):
    results.append((match.group('timestamp'), match.group('module'), match.group('message')))

return results
```

Example log data for testing

```
In [33]: test_log_data = '''
[2025-02-24 10:12:45] [ERROR] [Database_Module] Failed query from 192.168.0.1
[2025-02-24 11:30:22] [INFO] [User_Module] User profile updated.
[2025-02-24 12:45:00] [ERROR] [Payment_Module] Payment declined from 10.0.0.1
[2025-02-24 13:15:30] [ERROR] [Auth_Module] Invalid token access from 172.16.0.1
[2025-02-24 14:00:00] [WARN] [Network_Module] Slow response detected.
'''
```

Print output

```
In [34]: output = extract_critical_errors(test_log_data)
print(output)
```

```
[('2025-02-24 10:12:45', 'Database_Module', 'Failed query from 192.168.0.1. Error Code: 0x1234ABCD'), ('2025-02-24 12:45:00', 'Payment_Module', 'Payment declined from 10.0.0.200. Error Code: 0x87654321'), ('2025-02-24 13:15:30', 'Auth_Module', 'Invalid token access from 172.16.5.100. Error Code: 0xCAFEBA BE')]
```

This notebook was converted with convert.ploomber.io